EXCEL ASSIGNMENT 1

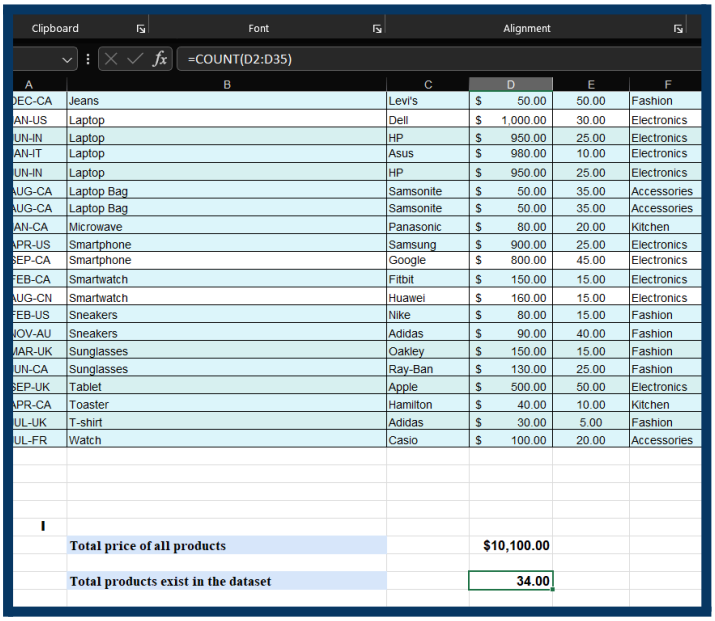
Data Exploration — Product Dataset

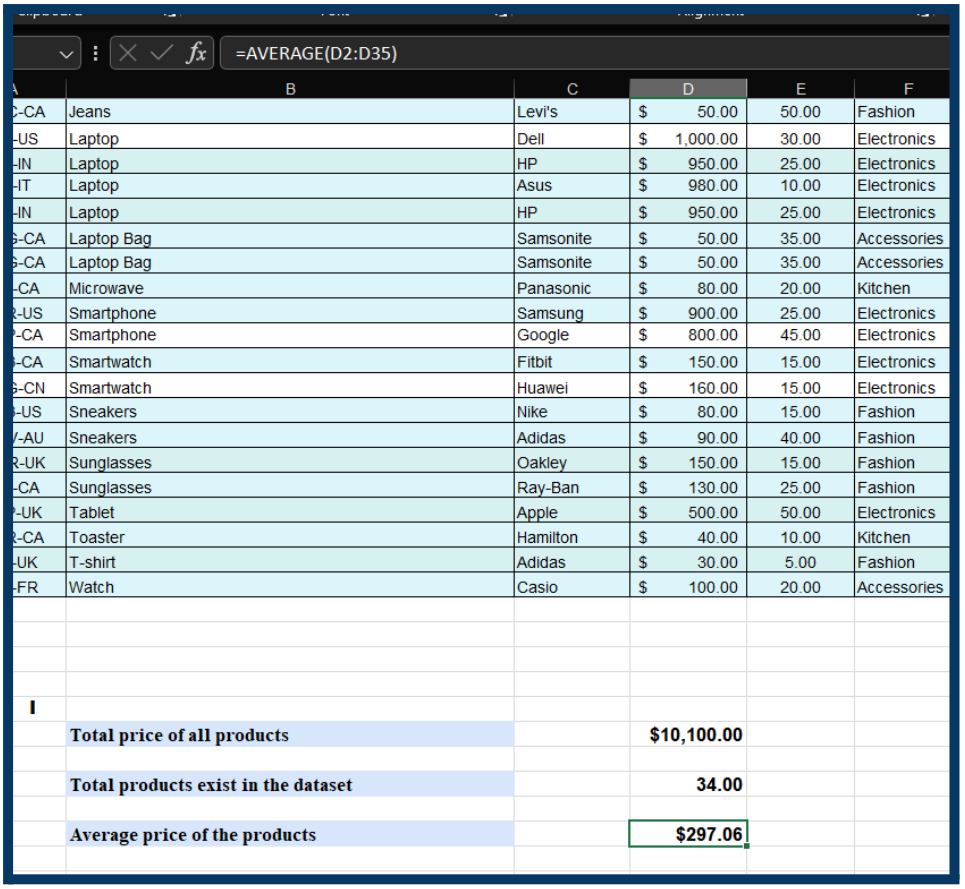
Project	Excel Data Exploration
Dataset size	34 product records
Functions covered	SUM, COUNT, AVERAGE, MIN, MAX, IF, SUMIF, COUNTIF, LEFT, RIGHT
Deliverables	Completed Excel workbook and formula evidence report

1. Basic Data Exploration

The SUM function calculates total price, COUNT counts numeric prices, and AVERAGE calculates the mean price.







=AVERAGE(D2:D35)

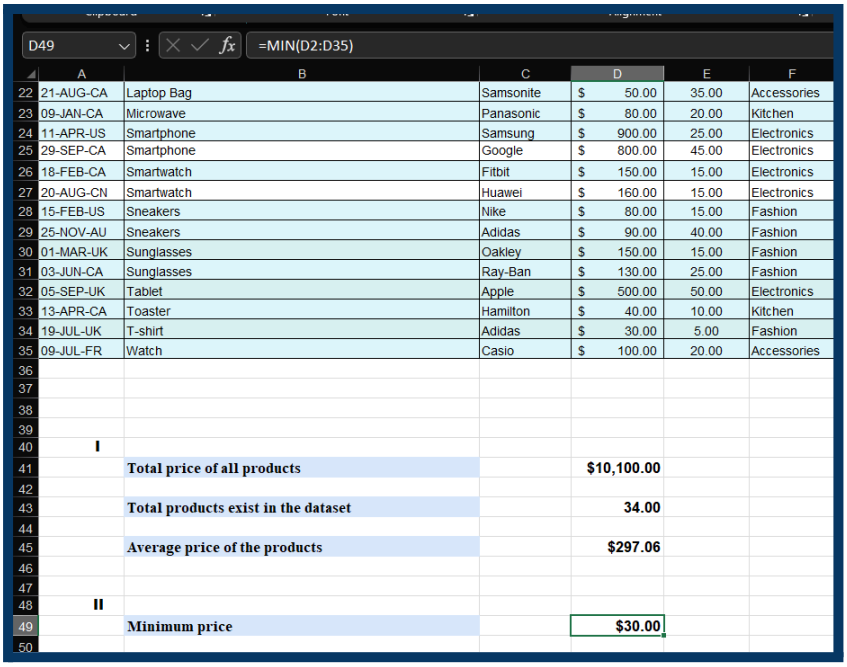
	B	C	D	E	F
-CA	Jeans	Levi's	$ 50.00	50.00	Fashion
-US	Laptop	Dell	$ 1,000.00	30.00	Electronics
-IN	Laptop	HP	$ 950.00	25.00	Electronics
-IT	Laptop	Asus	$ 980.00	10.00	Electronics
-IN	Laptop	HP	$ 950.00	25.00	Electronics
-CA	Laptop Bag	Samsonite	$ 50.00	35.00	Accessories
-CA	Laptop Bag	Samsonite	$ 50.00	35.00	Accessories
-CA	Microwave	Panasonic	$ 80.00	20.00	Kitchen
-US	Smartphone	Samsung	$ 900.00	25.00	Electronics
-CA	Smartphone	Google	$ 800.00	45.00	Electronics
-CA	Smartwatch	Fitbit	$ 150.00	15.00	Electronics
-CN	Smartwatch	Huawei	$ 160.00	15.00	Electronics
-US	Sneakers	Nike	$ 80.00	15.00	Fashion
-AU	Sneakers	Adidas	$ 90.00	40.00	Fashion
-UK	Sunglasses	Oakley	$ 150.00	15.00	Fashion
-CA	Sunglasses	Ray-Ban	$ 130.00	25.00	Fashion
-UK	Tablet	Apple	$ 500.00	50.00	Electronics
-CA	Toaster	Hamilton	$ 40.00	10.00	Kitchen
-UK	T-shirt	Adidas	$ 30.00	5.00	Fashion
-FR	Watch	Casio	$ 100.00	20.00	Accessories

I

Total price of all products	**$10,100.00**
Total products exist in the dataset	**34.00**
Average price of the products	**$297.06**

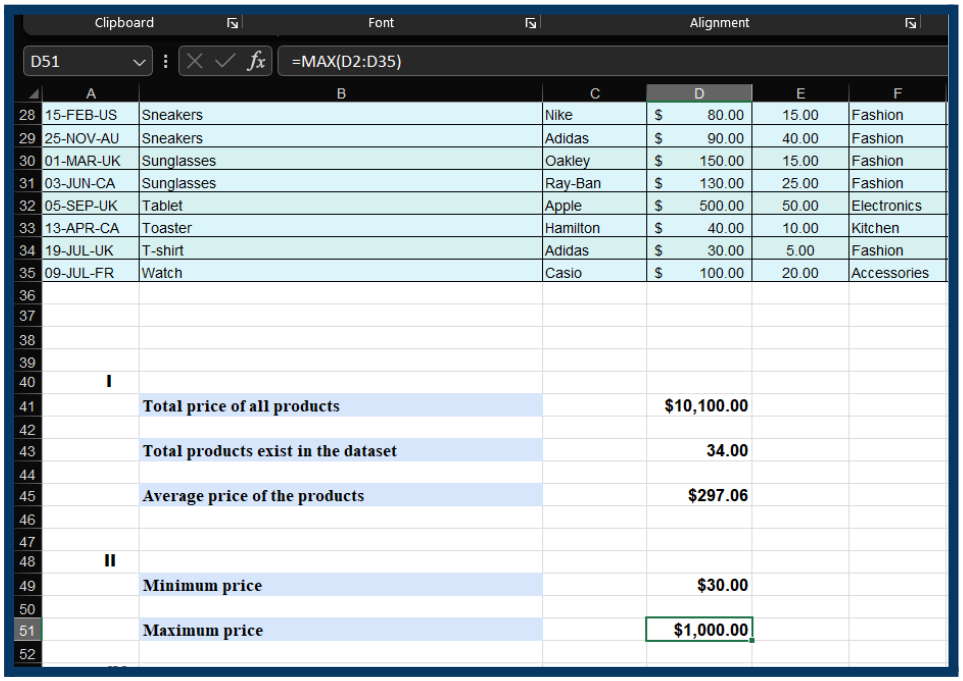
2. Minimum and Maximum

MIN returns the lowest price and MAX returns the highest price in the Price column.



D49 =MIN(D2:D35)

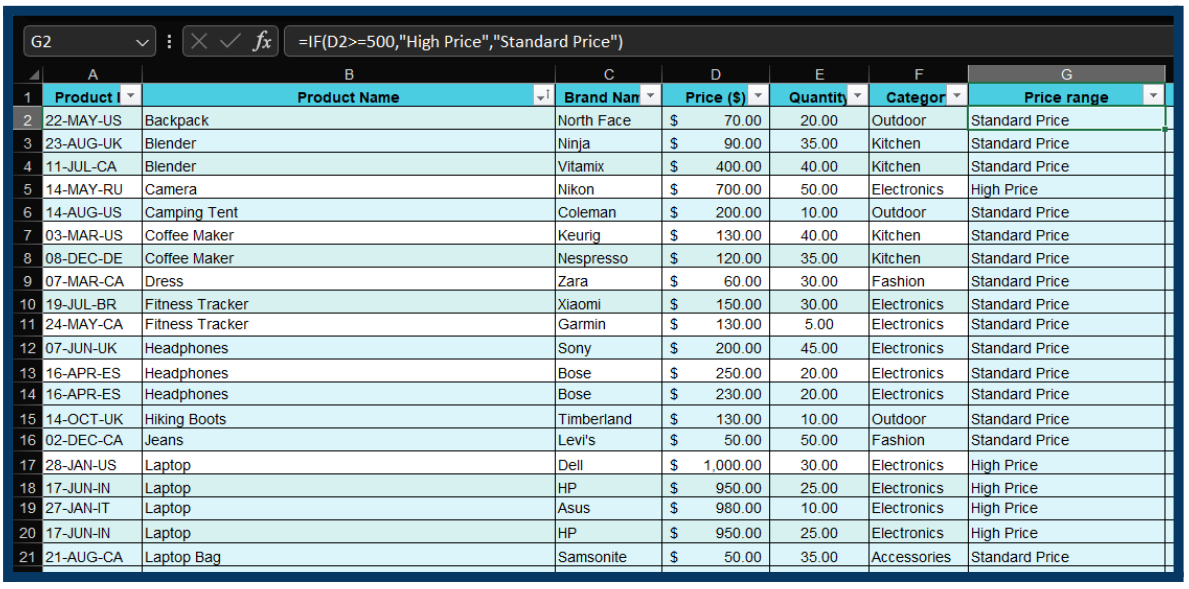
	A	B	C	D	E	F
22	21-AUG-CA	Laptop Bag	Samsonite	$ 50.00	35.00	Accessories
23	09-JAN-CA	Microwave	Panasonic	$ 80.00	20.00	Kitchen
24	11-APR-US	Smartphone	Samsung	$ 900.00	25.00	Electronics
25	29-SEP-CA	Smartphone	Google	$ 800.00	45.00	Electronics
26	18-FEB-CA	Smartwatch	Fitbit	$ 150.00	15.00	Electronics
27	20-AUG-CN	Smartwatch	Huawei	$ 160.00	15.00	Electronics
28	15-FEB-US	Sneakers	Nike	$ 80.00	15.00	Fashion
29	25-NOV-AU	Sneakers	Adidas	$ 90.00	40.00	Fashion
30	01-MAR-UK	Sunglasses	Oakley	$ 150.00	15.00	Fashion
31	03-JUN-CA	Sunglasses	Ray-Ban	$ 130.00	25.00	Fashion
32	05-SEP-UK	Tablet	Apple	$ 500.00	50.00	Electronics
33	13-APR-CA	Toaster	Hamilton	$ 40.00	10.00	Kitchen
34	19-JUL-UK	T-shirt	Adidas	$ 30.00	5.00	Fashion
35	09-JUL-FR	Watch	Casio	$ 100.00	20.00	Accessories
36						
37						
38						
39						
40	I					
41		**Total price of all products**		**$10,100.00**		
42						
43		**Total products exist in the dataset**		**34.00**		
44						
45		**Average price of the products**		**$297.06**		
46						
47						
48	II					
49		**Minimum price**		**$30.00**		
50						



	A	B	C	D	E	F
28	15-FEB-US	Sneakers	Nike	$ 80.00	15.00	Fashion
29	25-NOV-AU	Sneakers	Adidas	$ 90.00	40.00	Fashion
30	01-MAR-UK	Sunglasses	Oakley	$ 150.00	15.00	Fashion
31	03-JUN-CA	Sunglasses	Ray-Ban	$ 130.00	25.00	Fashion
32	05-SEP-UK	Tablet	Apple	$ 500.00	50.00	Electronics
33	13-APR-CA	Toaster	Hamilton	$ 40.00	10.00	Kitchen
34	19-JUL-UK	T-shirt	Adidas	$ 30.00	5.00	Fashion
35	09-JUL-FR	Watch	Casio	$ 100.00	20.00	Accessories
36						
37						
38						
39						
40	I					
41		**Total price of all products**		**$10,100.00**		
42						
43		**Total products exist in the dataset**		**34.00**		
44						
45		**Average price of the products**		**$297.06**		
46						
47						
48	II					
49		**Minimum price**		**$30.00**		
50						
51		**Maximum price**		**$1,000.00**		
52						

3. Logical Function — IF

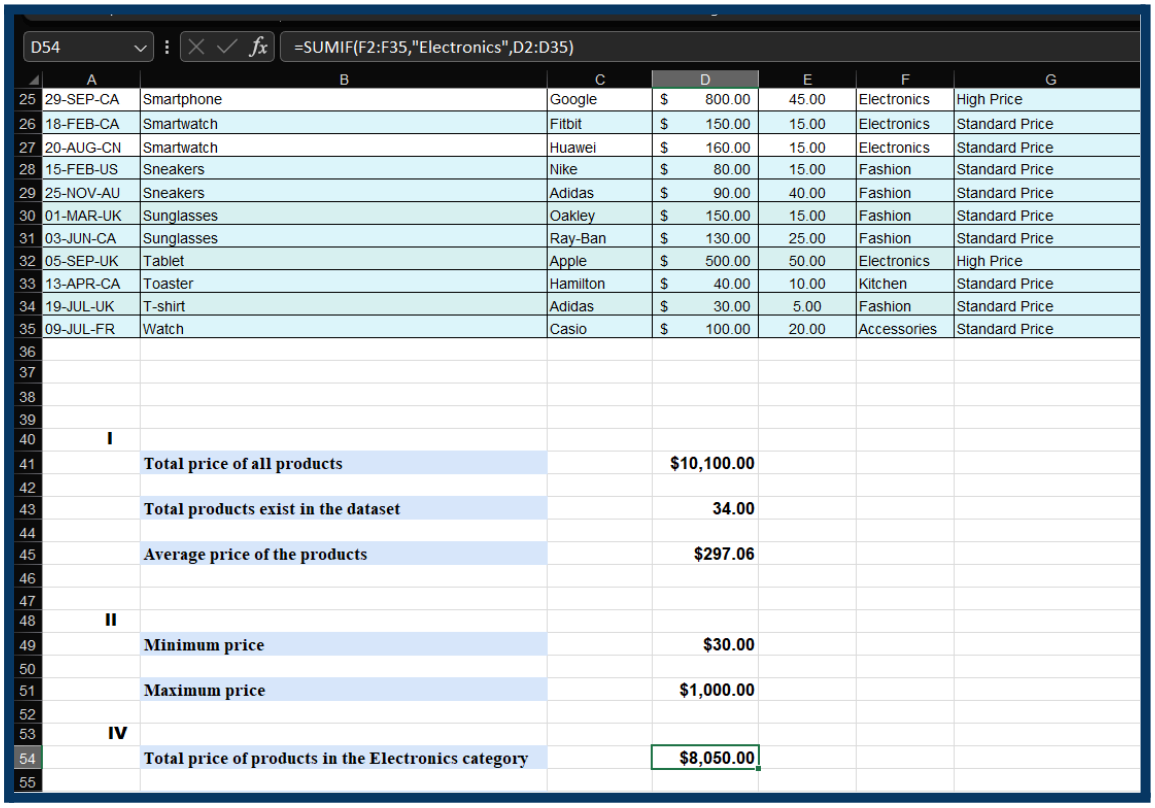
Price ³ $500 is classified as High Price; Price < $500 is classified as Standard Price.



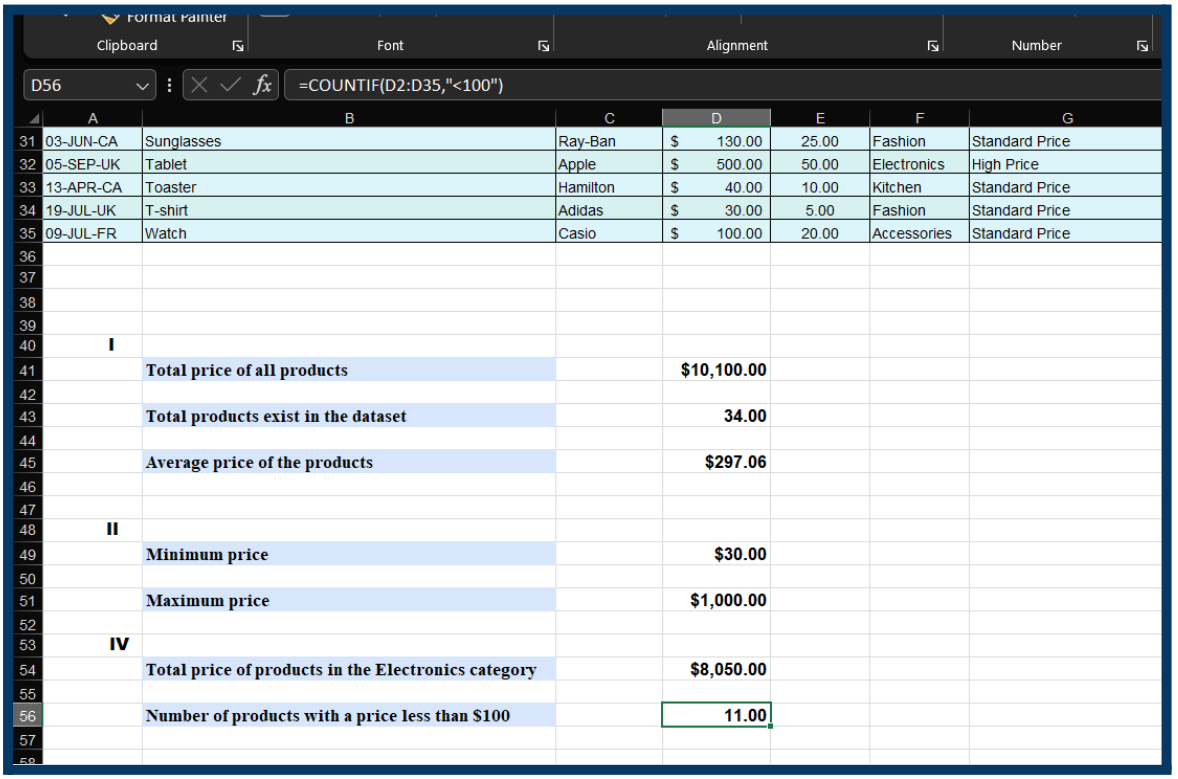
	A	B	C	D	E	F	G
1	Product I	Product Name	Brand Nan	Price ($)	Quantity	Categor	Price range
2	22-MAY-US	Backpack	North Face	$ 70.00	20.00	Outdoor	Standard Price
3	23-AUG-UK	Blender	Ninja	$ 90.00	35.00	Kitchen	Standard Price
4	11-JUL-CA	Blender	Vitamix	$ 400.00	40.00	Kitchen	Standard Price
5	14-MAY-RU	Camera	Nikon	$ 700.00	50.00	Electronics	High Price
6	14-AUG-US	Camping Tent	Coleman	$ 200.00	10.00	Outdoor	Standard Price
7	03-MAR-US	Coffee Maker	Keurig	$ 130.00	40.00	Kitchen	Standard Price
8	08-DEC-DE	Coffee Maker	Nespresso	$ 120.00	35.00	Kitchen	Standard Price
9	07-MAR-CA	Dress	Zara	$ 60.00	30.00	Fashion	Standard Price
10	19-JUL-BR	Fitness Tracker	Xiaomi	$ 150.00	30.00	Electronics	Standard Price
11	24-MAY-CA	Fitness Tracker	Garmin	$ 130.00	5.00	Electronics	Standard Price
12	07-JUN-UK	Headphones	Sony	$ 200.00	45.00	Electronics	Standard Price
13	16-APR-ES	Headphones	Bose	$ 250.00	20.00	Electronics	Standard Price
14	16-APR-ES	Headphones	Bose	$ 230.00	20.00	Electronics	Standard Price
15	14-OCT-UK	Hiking Boots	Timberland	$ 130.00	10.00	Outdoor	Standard Price
16	02-DEC-CA	Jeans	Levi's	$ 50.00	50.00	Fashion	Standard Price
17	28-JAN-US	Laptop	Dell	$ 1,000.00	30.00	Electronics	High Price
18	17-JUN-IN	Laptop	HP	$ 950.00	25.00	Electronics	High Price
19	27-JAN-IT	Laptop	Asus	$ 980.00	10.00	Electronics	High Price
20	17-JUN-IN	Laptop	HP	$ 950.00	25.00	Electronics	High Price
21	21-AUG-CA	Laptop Bag	Samsonite	$ 50.00	35.00	Accessories	Standard Price

4. Conditional Functions — SUMIF and COUNTIF

SUMIF totals prices for products in the Electronics category. COUNTIF counts products with prices below $100.



	A	B	C	D	E	F	G
25	29-SEP-CA	Smartphone	Google	$ 800.00	45.00	Electronics	High Price
26	18-FEB-CA	Smartwatch	Fitbit	$ 150.00	15.00	Electronics	Standard Price
27	20-AUG-CN	Smartwatch	Huawei	$ 160.00	15.00	Electronics	Standard Price
28	15-FEB-US	Sneakers	Nike	$ 80.00	15.00	Fashion	Standard Price
29	25-NOV-AU	Sneakers	Adidas	$ 90.00	40.00	Fashion	Standard Price
30	01-MAR-UK	Sunglasses	Oakley	$ 150.00	15.00	Fashion	Standard Price
31	03-JUN-CA	Sunglasses	Ray-Ban	$ 130.00	25.00	Fashion	Standard Price
32	05-SEP-UK	Tablet	Apple	$ 500.00	50.00	Electronics	High Price
33	13-APR-CA	Toaster	Hamilton	$ 40.00	10.00	Kitchen	Standard Price
34	19-JUL-UK	T-shirt	Adidas	$ 30.00	5.00	Fashion	Standard Price
35	09-JUL-FR	Watch	Casio	$ 100.00	20.00	Accessories	Standard Price
36							
37							
38							
39							
40	I						
41		**Total price of all products**		**$10,100.00**			
42							
43		**Total products exist in the dataset**		**34.00**			
44							
45		**Average price of the products**		**$297.06**			
46							
47							
48	II						
49		**Minimum price**		**$30.00**			
50							
51		**Maximum price**		**$1,000.00**			
52							
53	IV						
54		**Total price of products in the Electronics category**		**$8,050.00**			
55							



	A	B	C	D	E	F	G
31	03-JUN-CA	Sunglasses	Ray-Ban	$ 130.00	25.00	Fashion	Standard Price
32	05-SEP-UK	Tablet	Apple	$ 500.00	50.00	Electronics	High Price
33	13-APR-CA	Toaster	Hamilton	$ 40.00	10.00	Kitchen	Standard Price
34	19-JUL-UK	T-shirt	Adidas	$ 30.00	5.00	Fashion	Standard Price
35	09-JUL-FR	Watch	Casio	$ 100.00	20.00	Accessories	Standard Price
36							
37							
38							
39							
40	I						
41		**Total price of all products**		**$10,100.00**			
42							
43		**Total products exist in the dataset**		**34.00**			
44							
45		**Average price of the products**		**$297.06**			
46							
47							
48	II						
49		**Minimum price**		**$30.00**			
50							
51		**Maximum price**		**$1,000.00**			
52							
53	IV						
54		**Total price of products in the Electronics category**		**$8,050.00**			
55							
56		**Number of products with a price less than $100**		**11.00**			
57							

5. Text Processing — LEFT, RIGHT and MID

The Product ID format in the supplied data is DD-MMM-CC. LEFT extracts the first two characters (Day), RIGHT extracts the final two characters (Country Code), and MID extracts three characters starting at position 4 (Month).



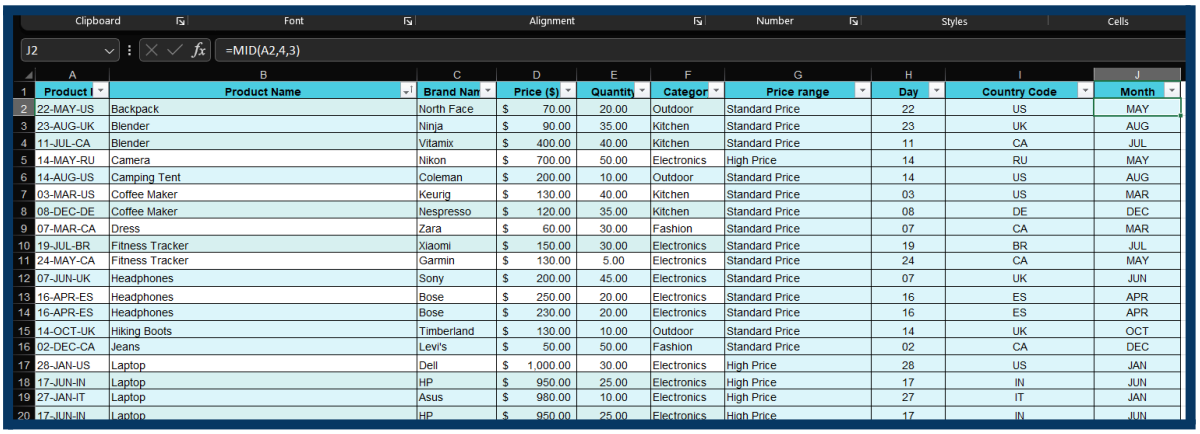
H2 =LEFT(A2,2)

	A	B	C	D	E	F	G	H
1	Product I	Product Name	Brand Nam	Price ($)	Quantity	Categor	Price range	Day
2	22-MAY-US	Backpack	North Face	$ 70.00	20.00	Outdoor	Standard Price	22
3	23-AUG-UK	Blender	Ninja	$ 90.00	35.00	Kitchen	Standard Price	23
4	11-JUL-CA	Blender	Vitamix	$ 400.00	40.00	Kitchen	Standard Price	11
5	14-MAY-RU	Camera	Nikon	$ 700.00	50.00	Electronics	High Price	14
6	14-AUG-US	Camping Tent	Coleman	$ 200.00	10.00	Outdoor	Standard Price	14
7	03-MAR-US	Coffee Maker	Keurig	$ 130.00	40.00	Kitchen	Standard Price	03
8	08-DEC-DE	Coffee Maker	Nespresso	$ 120.00	35.00	Kitchen	Standard Price	08
9	07-MAR-CA	Dress	Zara	$ 60.00	30.00	Fashion	Standard Price	07
10	19-JUL-BR	Fitness Tracker	Xiaomi	$ 150.00	30.00	Electronics	Standard Price	19
11	24-MAY-CA	Fitness Tracker	Garmin	$ 130.00	5.00	Electronics	Standard Price	24
12	07-JUN-UK	Headphones	Sony	$ 200.00	45.00	Electronics	Standard Price	07



I2 =RIGHT(A2,2)

	A	B	C	D	E	F	G	H	I
1	Product I	Product Name	Brand Nam	Price ($)	Quantity	Categor	Price range	Day	Country Code
2	22-MAY-US	Backpack	North Face	$ 70.00	20.00	Outdoor	Standard Price	22	US
3	23-AUG-UK	Blender	Ninja	$ 90.00	35.00	Kitchen	Standard Price	23	UK
4	11-JUL-CA	Blender	Vitamix	$ 400.00	40.00	Kitchen	Standard Price	11	CA
5	14-MAY-RU	Camera	Nikon	$ 700.00	50.00	Electronics	High Price	14	RU
6	14-AUG-US	Camping Tent	Coleman	$ 200.00	10.00	Outdoor	Standard Price	14	US
7	03-MAR-US	Coffee Maker	Keurig	$ 130.00	40.00	Kitchen	Standard Price	03	US
8	08-DEC-DE	Coffee Maker	Nespresso	$ 120.00	35.00	Kitchen	Standard Price	08	DE
9	07-MAR-CA	Dress	Zara	$ 60.00	30.00	Fashion	Standard Price	07	CA
10	19-JUL-BR	Fitness Tracker	Xiaomi	$ 150.00	30.00	Electronics	Standard Price	19	BR
11	24-MAY-CA	Fitness Tracker	Garmin	$ 130.00	5.00	Electronics	Standard Price	24	CA
12	07-JUN-UK	Headphones	Sony	$ 200.00	45.00	Electronics	Standard Price	07	UK
13	16-APR-ES	Headphones	Bose	$ 250.00	20.00	Electronics	Standard Price	16	ES
14	16-APR-ES	Headphones	Bose	$ 230.00	20.00	Electronics	Standard Price	16	ES



J2 =MID(A2,4,3)

	A	B	C	D	E	F	G	H	I	J
1	Product I	Product Name	Brand Nam	Price ($)	Quantity	Categor	Price range	Day	Country Code	Month
2	22-MAY-US	Backpack	North Face	$ 70.00	20.00	Outdoor	Standard Price	22	US	MAY
3	23-AUG-UK	Blender	Ninja	$ 90.00	35.00	Kitchen	Standard Price	23	UK	AUG
4	11-JUL-CA	Blender	Vitamix	$ 400.00	40.00	Kitchen	Standard Price	11	CA	JUL
5	14-MAY-RU	Camera	Nikon	$ 700.00	50.00	Electronics	High Price	14	RU	MAY
6	14-AUG-US	Camping Tent	Coleman	$ 200.00	10.00	Outdoor	Standard Price	14	US	AUG
7	03-MAR-US	Coffee Maker	Keurig	$ 130.00	40.00	Kitchen	Standard Price	03	US	MAR
8	08-DEC-DE	Coffee Maker	Nespresso	$ 120.00	35.00	Kitchen	Standard Price	08	DE	DEC
9	07-MAR-CA	Dress	Zara	$ 60.00	30.00	Fashion	Standard Price	07	CA	MAR
10	19-JUL-BR	Fitness Tracker	Xiaomi	$ 150.00	30.00	Electronics	Standard Price	19	BR	JUL
11	24-MAY-CA	Fitness Tracker	Garmin	$ 130.00	5.00	Electronics	Standard Price	24	CA	MAY
12	07-JUN-UK	Headphones	Sony	$ 200.00	45.00	Electronics	Standard Price	07	UK	JUN
13	16-APR-ES	Headphones	Bose	$ 250.00	20.00	Electronics	Standard Price	16	ES	APR
14	16-APR-ES	Headphones	Bose	$ 230.00	20.00	Electronics	Standard Price	16	ES	APR
15	14-OCT-UK	Hiking Boots	Timberland	$ 130.00	10.00	Outdoor	Standard Price	14	UK	OCT
16	02-DEC-CA	Jeans	Levi's	$ 50.00	50.00	Fashion	Standard Price	02	CA	DEC
17	28-JAN-US	Laptop	Dell	$ 1,000.00	30.00	Electronics	High Price	28	US	JAN
18	17-JUN-IN	Laptop	HP	$ 950.00	25.00	Electronics	High Price	17	IN	JUN
19	27-JAN-IT	Laptop	Asus	$ 980.00	10.00	Electronics	High Price	27	IT	JAN
20	17-JUN-IN	Laptop	HP	$ 950.00	25.00	Electronics	High Price	17	IN	JUN